YASHWANTH GOWDA S N

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OBJECTIVE

Computer & Communication Engineering student focused on Backend Engineering, Distributed Systems, Cloud Computing, and Edge AI, seeking Software Engineering Internship opportunities.

SKILLS

Languages	Java, Python, C, C++, SQL, JavaScript
Backend	Spring Boot, Spring Cloud Gateway, Spring Security, FastAPI, REST APIs, WebSockets, CDC Pipelines
Frontend	React.js, HTML, CSS
Databases	MySQL, MongoDB, Redis, ChromaDB
AI / ML	TensorFlow, TensorFlow Lite Micro, INT8 Quantization, LangChain, RAG, Gemini API, OpenAI API, Ollama
Embedded / IoT	ESP32, NXP FRDM-MCXN236, ESP-NOW, UART, SPI, Arduino
DevOps	Docker, Kubernetes, Git, Linux
Core CS	DSA, OOP, DBMS, OS, Computer Networks, Distributed Systems
Tools	Maven, Gradle, Postman, IntelliJ IDEA, VS Code

PROJECTS

RESq Mesh – Edge AI & Autonomous Hardware Telemetry Mesh GitHub

Built an edge network-health prediction system using an NXP FRDM-MCXN236 (Cortex-M33) and an INT8-quantized TensorFlow Lite Micro model. Achieved 10.3 ms on-device inference latency with a 13,572-byte tensor arena. Integrated ESP32-based peer-to-peer forwarding over ESP-NOW and designed offline state synchronization using causal sequence tracking and HMAC-SHA256 integrity checks.

Tech Stack: C/C++, Python, TFLite Micro, INT8 Quantization, NXP MCXN236, ESP32, ESP-NOW, HMAC-SHA256

Cloud Native API Gateway

Developed a cloud-native API gateway using Spring Cloud Gateway with JWT authentication, Redis-based rate limiting, Docker containerization, and Kubernetes-ready deployment. Integrated Kafka for asynchronous communication and Resilience4j for fault tolerance.

Tech Stack: Java, Spring Boot, Spring Cloud Gateway, Redis, Kafka, Docker, Kubernetes, JWT, Resilience4j

Persistent Memory AI Assistant

Built an AI assistant using LangChain, ChromaDB, and RAG, implementing semantic search and contextual memory retrieval for multi-session conversations.

Tech Stack: Python, FastAPI, LangChain, ChromaDB, RAG, Gemini API, OpenAI API, Docker

EcoSync AI – 3rd Place, EcoSpark 2026

Built an AI-powered IoT composting platform using ESP32 sensors, Spring Boot, and React for real-time monitoring and AI-assisted compost analysis. Secured 3rd Place at EcoSpark 2026.

Tech Stack: ESP32, Arduino, C++, React.js, Spring Boot, MySQL, Gemini API

ACHIEVEMENTS

- Secured **1st Place – Overall Winner** at **MLH Hacktoberfest Hack Day Bengaluru 2026**, winning 10,000 cash and an Arduino UNO Q development board and ARMORIQ merchandise and additional sponsor rewards.
- Secured **1st Place** in the ICARUS – CanSat Telemetry Analysis Competition at RV University.
- Secured **2nd Place** in GitSetGo!, an open-source Git/GitHub relay race at RV University.
- Received the **Suzlon Young Brilliance Award 2026**.
- Secured **3rd Place** at the EcoSpark 2026 Ideathon.
- Ranked among the **Top 10 Teams** at GDG TechSprint 2025.
- Solved **250+ LeetCode problems**; achieved an **Apprentice rating of 1587** in contests.

EDUCATION

B.E. Computer & Communication Engineering **Expected May 2029**

KS Institute of Technology (KSIT), Bengaluru

CGPA: **8.52/10.0**